

L-2-1: Useful Statistical Procedures for Nowcasting, Part 1: Time Series Analysis

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What is a Time-Series?

- A time-series is an array of numbers that can indexed by the date of the observation or its relative position in the array.
- For example, quarterly values of real GDP for the USA are shown in the Excel spreadsheet on the right.
- The first data point has a time-index of 1947Q1.

	A	B	C
1	Date	Real GDP	
2	1947Q1	2034.5	
3	1947Q2	2029	
4	1947Q3	2024.8	
5	1947Q4	2056.5	
6	1948Q1	2087.4	
7	1948Q2	2121.9	
8	1948Q3	2134.1	
9	1948Q4	2136.4	
10	1949Q1	2107	
11	1949Q2	2099.8	
12	1949Q3	2121.5	
13	1949Q4	2103.7	
14	1950Q1	2186.4	
15	1950Q2	2253	
16	1950Q3	2340.1	
17	1950Q4	2384.9	
18	1951Q1	2417.3	
19	1951Q2	2459.2	
20	1951Q3	2509.9	
21	1951Q4	2515.4	
22	1952Q1	2542.3	
23	1952Q2	2547.8	
24	1952Q3	2566.2	
25	1952Q4	2650.4	

Definition of Time-Series

- Note that GDP("1947Q1") = 2034.5
- Since it is in the first position of the array, GDP(1) = 2034.5
- Likewise, GDP("1952Q1") = GDP(21) = 2542.3

	A	B	C
1	Date	Real GDP	
2	1947Q1	2034.5	
3	1947Q2	2029	
4	1947Q3	2024.8	
5	1947Q4	2056.5	
6	1948Q1	2087.4	
7	1948Q2	2121.9	
8	1948Q3	2134.1	
9	1948Q4	2136.4	
10	1949Q1	2107	
11	1949Q2	2099.8	
12	1949Q3	2121.5	
13	1949Q4	2103.7	
14	1950Q1	2186.4	
15	1950Q2	2253	
16	1950Q3	2340.1	
17	1950Q4	2384.9	
18	1951Q1	2417.3	
19	1951Q2	2459.2	
20	1951Q3	2509.9	
21	1951Q4	2515.4	
22	1952Q1	2542.3	
23	1952Q2	2547.8	
24	1952Q3	2566.2	
25	1952Q4	2650.4	

Key Attributes of a Time-Series

- A time-series is an array of numbers (usually double-precision) with attributes that determine how those observations will be managed in specific situations. Key time-series attributes include:
- *Default frequency.* Real GDP, for example, can be *quarterly*, indicating that there will be 4 observations per year. There are several time-series frequencies, including weekly, monthly, half-yearly, 5-day week ("financial"), and daily. There is even a time-stamped frequency that allows for minutes/seconds.
- *Default conversion method* – this is the mathematical procedure used to move/convert the series from a higher frequency to a lower frequency, for example from quarterly to annual. Standard methods include first, last, sum, maximum, minimum or the average of the values of the period. Also, one may optionally ignore missing values when converting to a lower frequency.

Key Attributes of a Time-Series

- *Default interpolation method* – the mathematical procedure used to move/interpolate the series from a lower frequency to a higher frequency, for example from annual to quarterly. Standard methods include constant, linear, cubic-spline, and the Chow-Lin interpolation. Note that the same methods can be used to replace missing values in a time-series without changing its frequency.

- *The position of the observation* in the period (i.e., beginning, end, mid-point). For example, for a quarterly series, the observation may be positioned in the first, second, or third month of the quarter. The position of the observation is important when moving from a low to a higher frequency.

Time-Series Matrix

- A time-series matrix is a group of N > 1 time-series presented at a common or target frequency (e.g., quarterly).

- Each series may have a default frequency that is different from the base frequency, for example, quarterly. Typically, the default conversion/interpolation method of the series is used to move the time-series to the target frequency before constructing the time-series matrix.

Time-Series Matrix

- Each series need not have the same date range. There could be missing values at the beginning or end of the time-series matrix.

- For statistical analysis, one often works with the 'common overlap' (defined next slide) of the series such that there are no missing values at the beginning nor end of the time-series matrix.

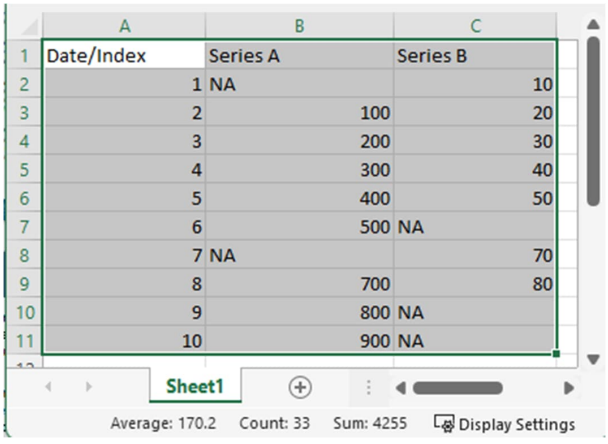
Time-Series Matrix with 2 Series

Consider the time-series matrix shown on the right. It has 2 series, each with 10 observations, some of which are missing (i.e., NA).

The common overlap involves the observations in rows 2 to 8 inclusive (missing values within the period are allowed, but not at the end or beginning of the period).

Notice that, starting from the first row (1) and working down, the first row that does not contain a missing value is row 2.

Starting from the last row (10) and working up, the last row that does not contain a missing value is row 8.



	A	B	C
1	Date/Index	Series A	Series B
2	1	NA	10
3	2	100	20
4	3	200	30
5	4	300	40
6	5	400	50
7	6	500	NA
8	7	NA	70
9	8	700	80
10	9	800	NA
11	10	900	NA

Multivariate statistical procedures would invariably base their analysis on the observations in rows 2 to 8 inclusive, unless some statistical procedure is used to replace the missing values in rows 1(Series A) and rows 9 and 10 (Series B).

Flow versus Stock Variables

- Time-series variables are often classified according to whether they are ‘flow’ or ‘stock/index’ variables.
- Flow variables are variables that start at zero at the beginning of each period (e.g., a year). The final value at the end of the period (e.g., a year) measures the amount of the item/good produced during a given period.
- For example, if a time-series is quarterly and a flow variable (e.g., real GDP), the value of real GDP would necessarily be zero at the beginning of the year (i.e., January 1). At the end of the year, December 31, the annual value of real GDP would be the sum of its 4 quarterly values for the year (i.e., total output for the year).

Flow versus Stock Variables

- By contrast, stock variables measure the amount of an item (e.g., the money supply) at a particular point in time. It will not be zero at the beginning of each period.
- Stock variables are normally positioned at the end of the period.
- Researchers typically use the sum or average of the higher frequency observations within the period to convert a flow variable from a higher to a lower frequency.
- There are far more options to consider in the case of flow variables. One can use the first observation of the period, the last, the mid-point, the average, maximum, minimum, or even the median. The position of the converted flow variable with the lower frequency time frame depends on the method used to convert the series to the lower frequency.

Frequency Conversion using EViews

Low to High Frequency Conversion (Interpolation)

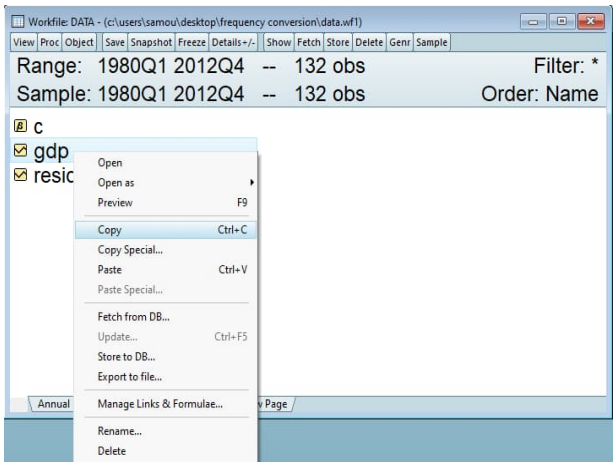
Frequency Conversion: Low to High

- It is easy in EViews to convert lower frequency data (e.g., quarterly frequency) to higher frequency data (e.g., monthly frequency).
- Suppose you have quarterly *GDP* data and wish to convert to monthly data.

Converting Quarterly Frequency to Monthly:

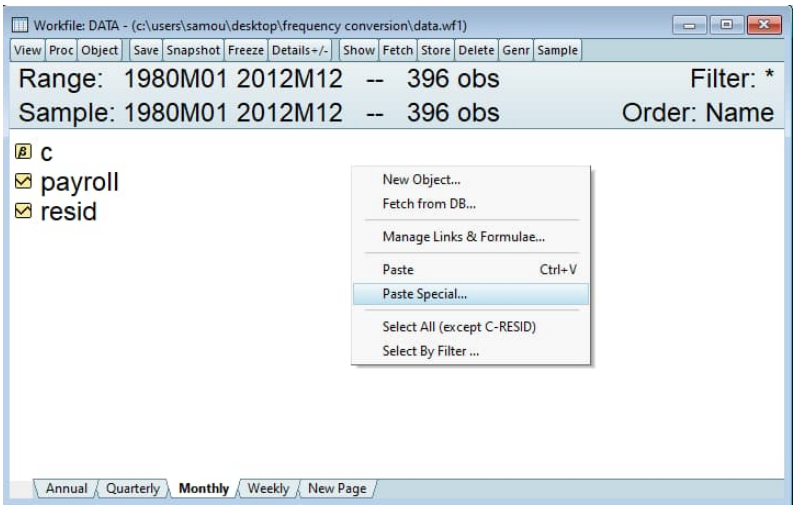
1. Open workfile *Data.wf1*. Click on the *Quarterly* page.
2. Right-click on *GDP* series and select Copy.

NOTE: The *Quarterly* page in this example is the source page, while the *Monthly* page is the destination page.



Frequency Conversion: Low to High (cont'd)

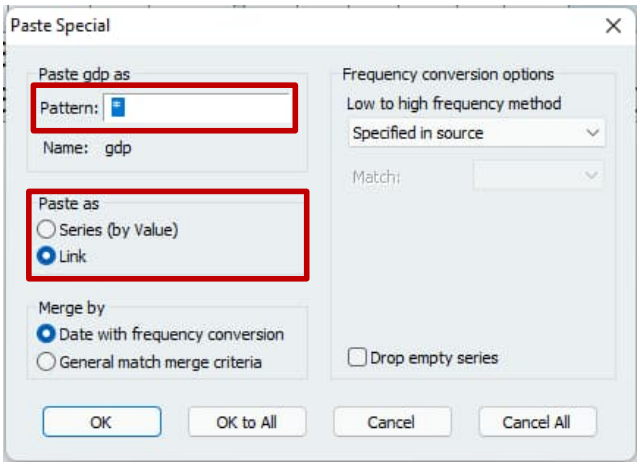
- Click on the *Monthly* page
- Click anywhere on the white workfile space
- Lastly, right-click and select Paste Special.



Frequency Conversion: Low to High (cont'd)

4. The *"Paste Special"* dialog box opens. This is a quick review of the main elements:

- ✓ "***Pattern***" allows you to change the name of the series.
- ✓ If you want to keep the same name, leave this area as is (with an "*" in it).
- ✓ "***Paste as***" allows you to paste as a "***Series (by Value)***" or as a "***Link***".
- ✓ If you paste as a **series**, then the copy will be a one-off event.
- ✓ If you paste as a link, the data in the ***Monthly*** page will be permanently linked to the data in the ***Quarterly*** page, so if the data in the source page (***Quarterly***) changes it will also change in the destination page (***Monthly***) .

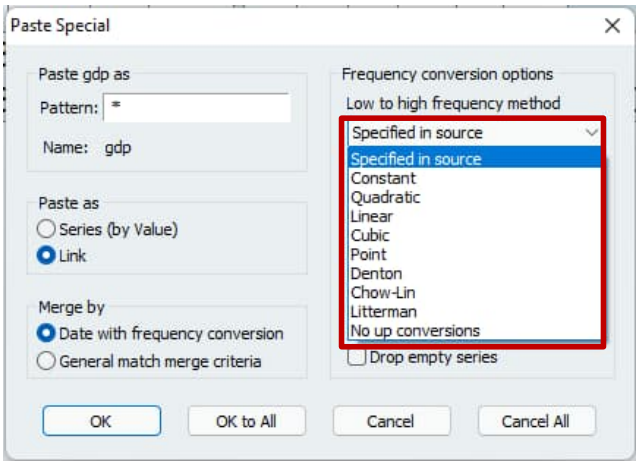


Method of Conversion: Low to High

- On the right-hand side there is the *Frequency Conversion Options"* section.
  - ✓There are two main options: "*High to low frequency method*" and "*Low to high*."
  - ✓Since we want to convert quarterly to monthly data, click the "*Low to high frequency method*" combo box/drop-down menu to select the type of interpolation method to be performed.

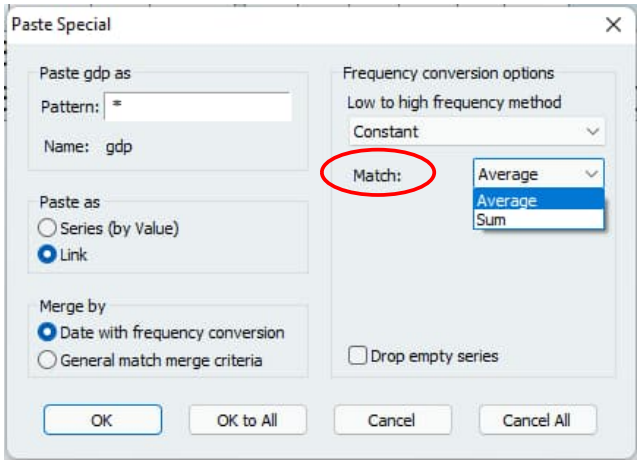
Low to high frequency method:

- ✓ ***Constant***
- ✓ ***Quadratic***
- ✓ ***Linear***
- ✓ ***Cubic Spline***
- ✓ ***Point***
- ✓ ***Denton***
- ✓ ***Chow-Lin***
- ✓ ***Litterman***



Method of Conversion: Low to High

Once the interpolation method is specified, you will need to specify the 'matching' method to use when moving to the higher frequency.



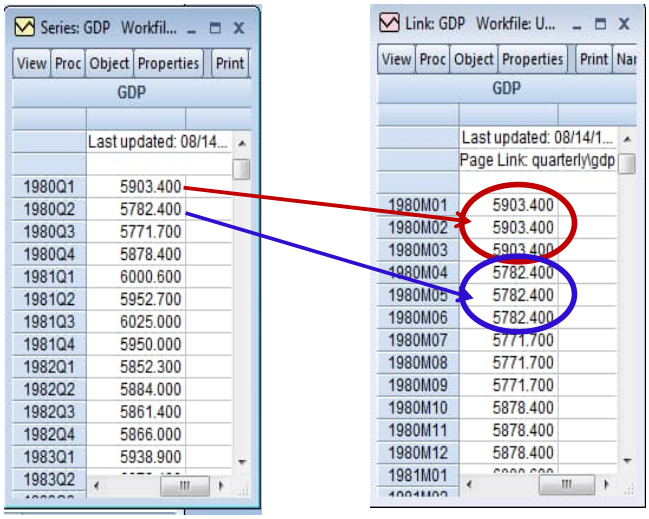
Method	Available Matching Methods
Constant	Average, Sum
Quadratic	Average, Sum
Linear	First, Last
Cubic	First, Last
Point	First, Last
Denton	First, Last, Average, Sum
Chow-lin	First, Last, Average, Sum
Litterman	First, Last, Average, Sum

Frequency Conversion: Low to High Example 1: Constant (Match: Average)

- ***Constant (Match: Average):*** repeats the value of the low frequency data for each of the high frequency observations associated with a particular low frequency period.

<u>Low to High Conversion: Example 1</u>

1. Click on the *Quarterly* page of the *Data.wf1* workfile.
2. Right-click on *GDP* series and select Copy.
3. Click on the *Monthly* page, right-click and select Paste Special.
4. On the *Paste Special* dialog box, select Link under *Paste as* section.
5. Select Date *with frequency conversion* under *Merge by* section.
6. Select Constant under "*Low to high frequency method.*"
7. Select Average for the "Match:" setting.
8. Click OK.



- This method repeats the low-frequency value (quarterly, in our case) for all the corresponding high-frequency observations for the selected low frequency data (i.e., all months in that quarter).

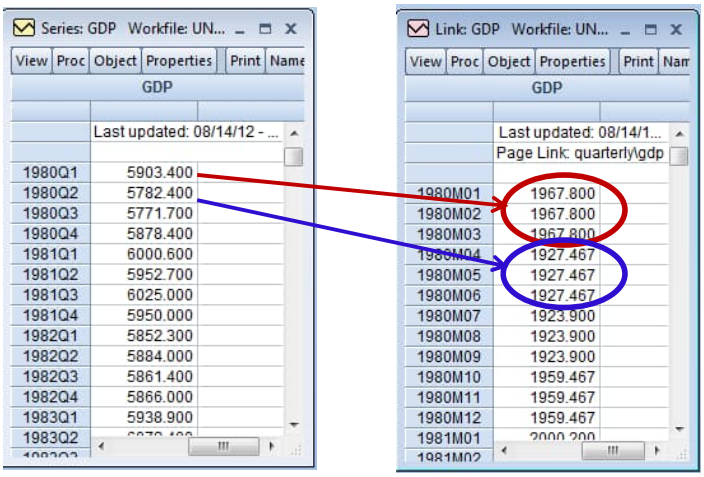
Frequency Conversion: Low to High Example 2: Constant (Match: Sum)

- *Constant (Match: Sum):* divides the low frequency value by the number of observations in the high frequency page that is associated with the low frequency period.

<u>Low to High Conversion: Example 2</u>

1. Follow steps 1-5 from Example 1.
2. Under *Pattern*, name this series *gdp1* (to not confuse it with the previous example).
3. On the *Paste Special* dialog box, select Constant under "*Low to high frequency method.*"
4. Select Sum for the "Match:" setting.
5. Click OK.

By contrast with the previous method, i.e., average, all observations in the high frequency series are filled with low frequency data divided by the number of periods in the high frequency page.



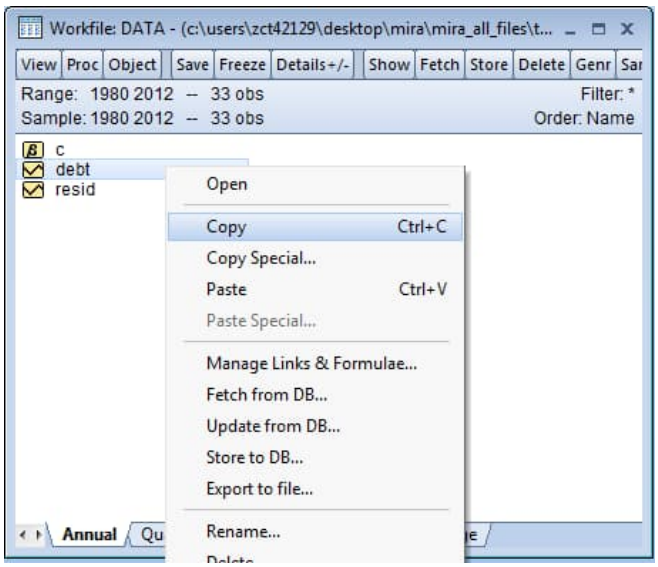
- In this example, each month in each quarter has the same value as the quarterly data divided by 3 (since there are 3 months in a quarter).

Frequency Conversion: Low to High Example 3: Quadratic (Match: Average)

- *Quadratic (Match: Average)*: performs a local quadratic interpolation of the low frequency data to fill in the high observations.
- As another example, let's convert the annual *debt* data from the *Annual* page into quarterly data (in the *Quarterly* page) using this conversion method.

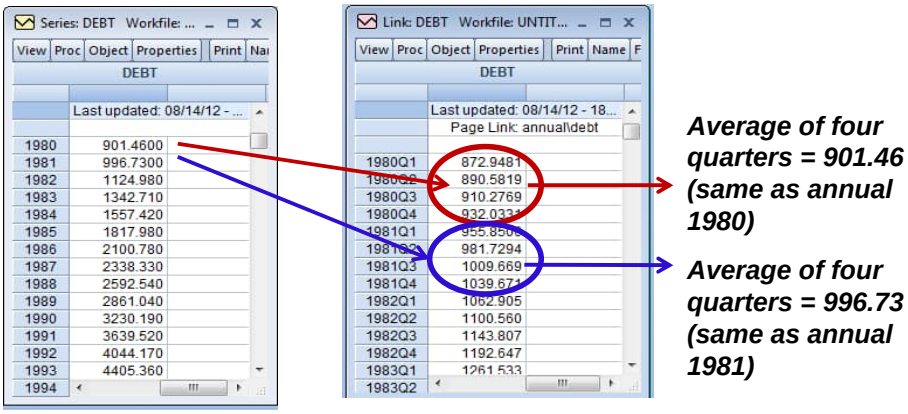
Low to High Conversion: Example 3

1. Click on the *Annual* page of the *Data.wf1* workfile. Right-click on *debt* series and select Copy.
2. Click on the *Quarterly* page, right-click and select Paste Special.
3. On the *Paste Special* dialog box, select Link under *Paste as* section.
4. Select Date *with frequency conversion* under *Merge by* section.
5. Select Quadratic under "*Low to high frequency method*".
6. Select Average for the "Match:" setting.
7. Click OK.



Frequency Conversion: Low to High Example 3: Quadratic-Match Average (cont'd)

- This method fits a local quadratic polynomial by taking sets of three adjacent points from the low frequency data (annual data in this example) and fitting a quadratic so that the average of the high frequency points matches the low frequency data observed.

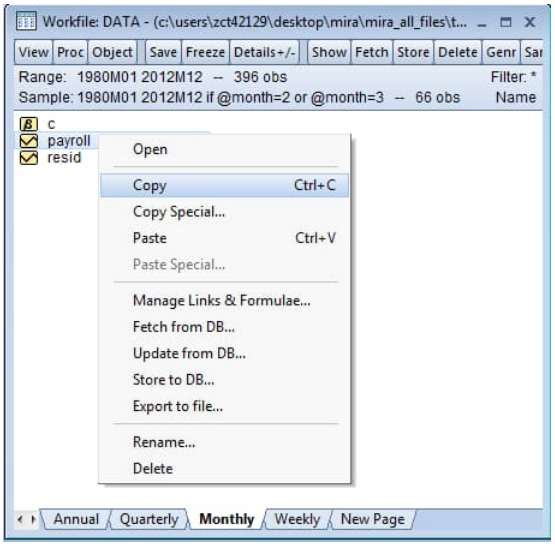


Frequency Conversion: Low to High
Example 4: Linear (Match: Last)

- *Linear (Match: Last)* inserts the low frequency observation into the last period of the high frequency data, then performs linear interpolation on the other values.
- As another example, consider the case of converting monthly *payroll* data into weekly data by using the *"Linear (Match: Average)"* conversion method.

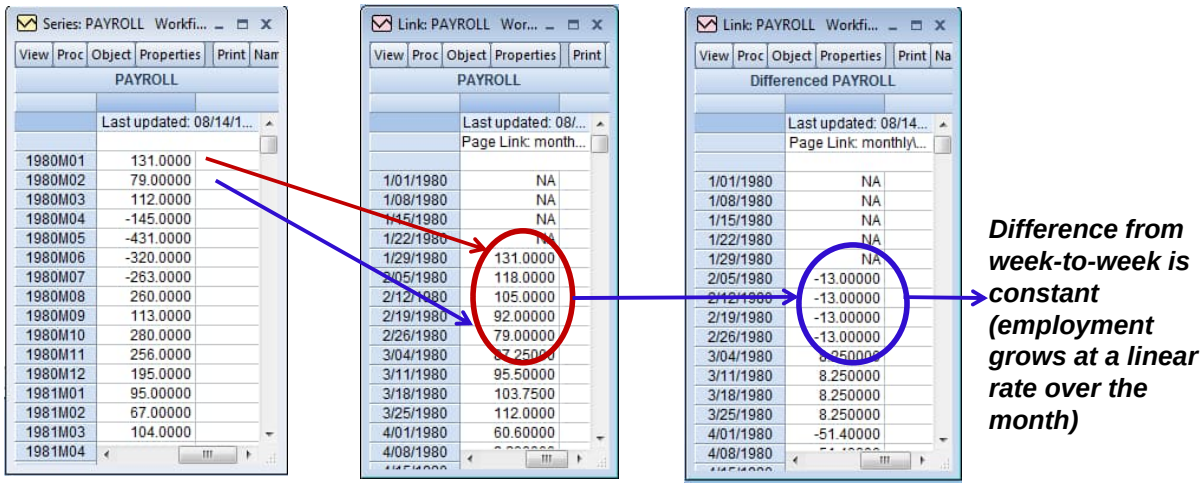
Low to High Conversion: Example 4

1. Click on the *Monthly* page of the *Data.wf1* workfile. Right-click on *payroll* series and select Copy.
2. Click on the *Weekly* page, right-click and select Paste Special.
3. On the *Paste Special* dialog box, select Link under *Paste as* section.
4. Select Date *with frequency conversion* under *Merge by* section.
5. Select Linear under "*Low to high frequency method*."
6. Select Last for the "Match:" setting.
7. Click OK.



Frequency Conversion: Low to High
Example 4: Linear (Match: Last) (cont'd)

- This method first places the monthly observation in the last weekly observation of the corresponding month. In-between weekly observations are filled by performing a linear interpolation between the last week of the previous month and the last week of the current month.



Difference from week-to-week is constant (employment grows at a linear rate over the month)

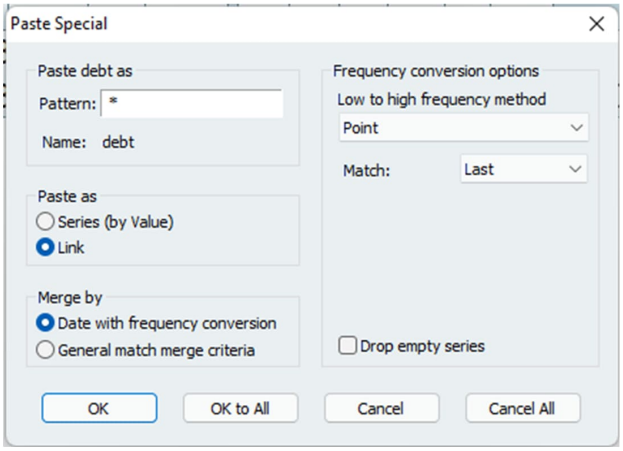
- Note that the linear (Match ***first***) option places the monthly observation in the ***first*** weekly position of the corresponding month.

Frequency Conversion: Low to High Example 5: Point (Match: Last)

- The point match last method simply copies the low frequency data into the last observation for the corresponding high frequency range.

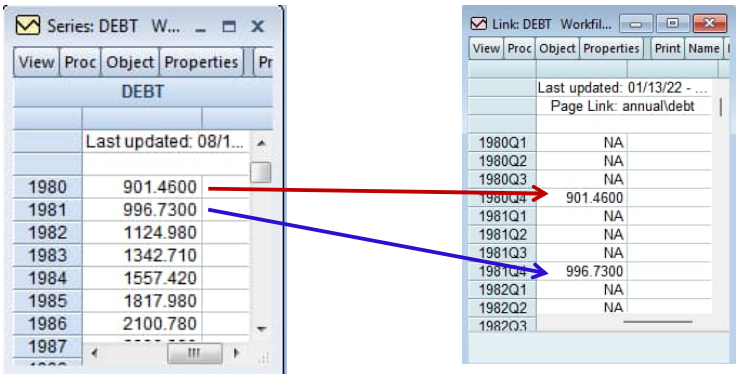
Low to High Conversion: Example 5 Point

1. Click on the *Annual* page. Right-click on *debt* series and select Copy.
2. Click on the *Quarterly* page, right-click and select Paste Special.
3. On the *Paste Special* dialog box, under *Pattern,* name this series *debt* (to not confuse it with an earlier example).
4. On the *Paste Special* dialog box, under *Paste as*, select Series (by Value).
5. For the low to high frequency method, select Point with a matching criterion of last.
6. Select Last for the "Match:" setting.
7. Click OK.



Frequency Conversion: Low to High Example 5 Point (Last) (cont'd)

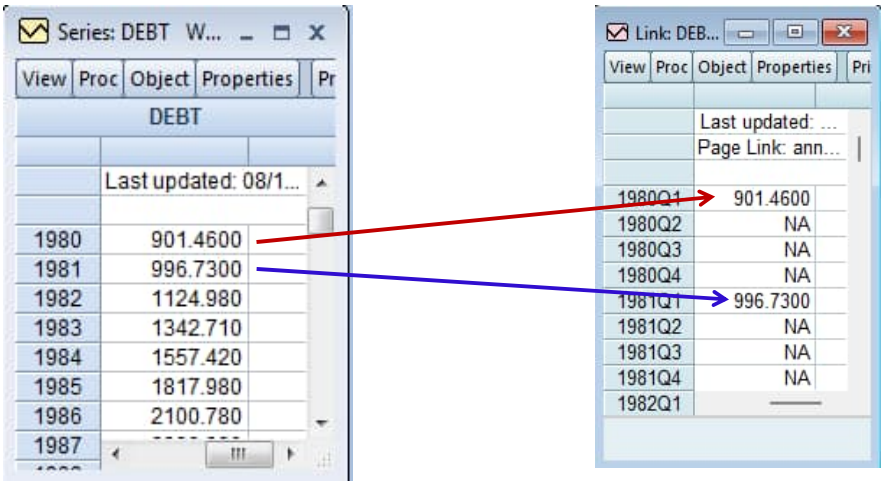
- The annual value is placed in Q4 of each year, while all the other quarters show "NA."



Also, selecting 'First' rather than 'Last' positions the low frequency value in the first period of the corresponding higher frequency period.

Frequency Conversion: Low to High Example 5 Point (Match: First) (cont'd)

- The annual value is placed in Q1 of each year, while all the other quarters show "NA."



Frequency Conversion: Low to High Advanced Methods

EViews also supports more advanced interpolation methods that involve the use of high frequency indicators (other time-series) to assist in estimating missing values in the series (if any).

These methods are called "Cublic-Spline", "Denton", "Chow-Lin", and "Litterman" in the Low-to-High frequency combo box.

These methods are beyond the scope of this course, but easy to implement following the same procedures explained above.

Please refer to the EViews manual for basic information on the implementation of these methods. You may also wish to read the original articles that develop/justify these methods. See the EViews manual.

Frequency Conversion: Low to High Chow-Lin Interpolation

The Chow-Lin Interpolation is a regression-based method that finds values of a series x by relating one or more high-frequency indicator Z series to a lower-frequency benchmark through the equation:

$$x(t) = Z(t)\beta + a(t)$$

where is β is a vector of coefficients and a(t) is a random variable with mean zero and covariance matrix V.

EViews uses a state space model with the following time series model for the states:

$$a(t) = \rho\, a(t-1) + \epsilon(t)$$

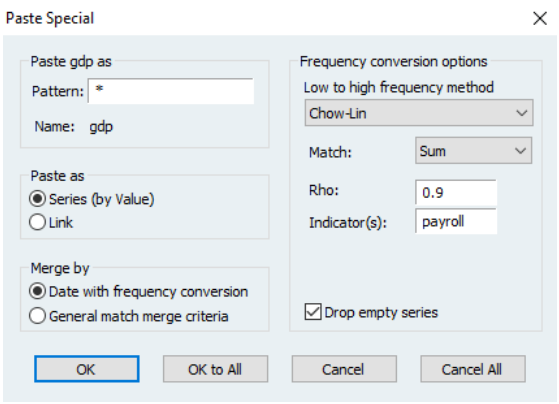
Where $\epsilon(t) \sim N(0, \sigma^2)$ and $|\rho| < 1$. The parameters ρ and β are estimated via maximum likelihood and the Kalman filter, and the interpolated series is subsequently calculated with Kalman smoothing.

Source: EViews 12 Users Guide I, pp 176
Chow, G. C., & Lin, A. L. (1971). "Best linear unbiased interpolation, distribution, and extrapolation of time series by related series," *The Review of Economics and Statistics*, 53(4), 372-375.

Frequency Conversion: Low to High Chow-Lin Interpolation

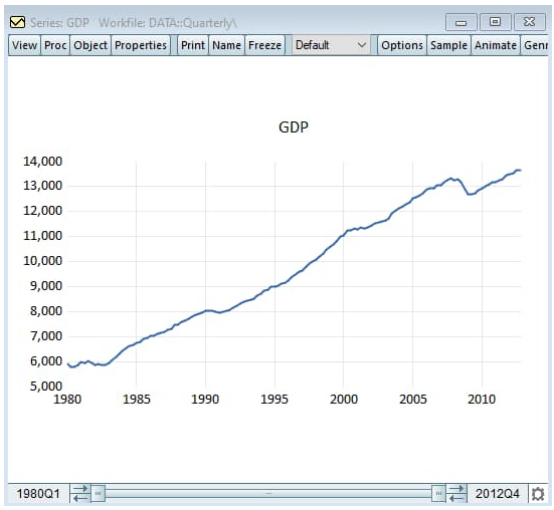
Exercise: Convert "gdp" series from Quarterly to Monthly frequency

- Rho (ρ) reflects the persistence in the error-term of the regression equation
- 0 ≤ ρ ≤ 1 (lower ρ will introduce more noise)
- Suggested procedure to choose ρ:
  - Start from a high value (e.g., 0.9) for macro data
  - Visually evaluate the interpolated series relative to the original
  - If necessary, move ρ down by 0.05 at a time and repeat the procedure
- Indicator(s) in the target frequency (high) can be added to the interpolation procedure

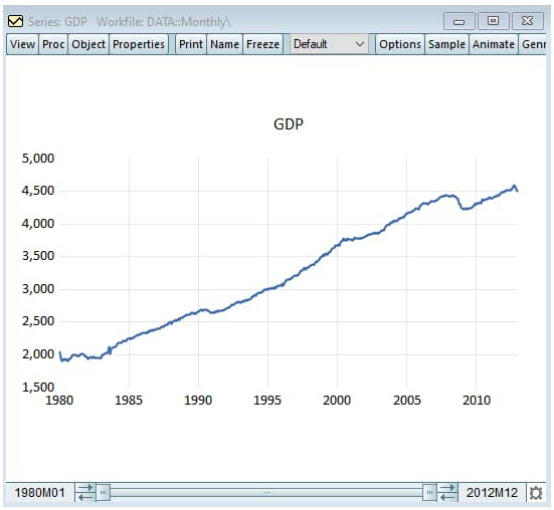


Frequency Conversion: Low to High Chow-Lin Interpolation

Quarterly



Monthly



Frequency Conversion using EViews

High to Low Frequency Conversion (Series Consolidation)

Frequency Conversion: High to Low

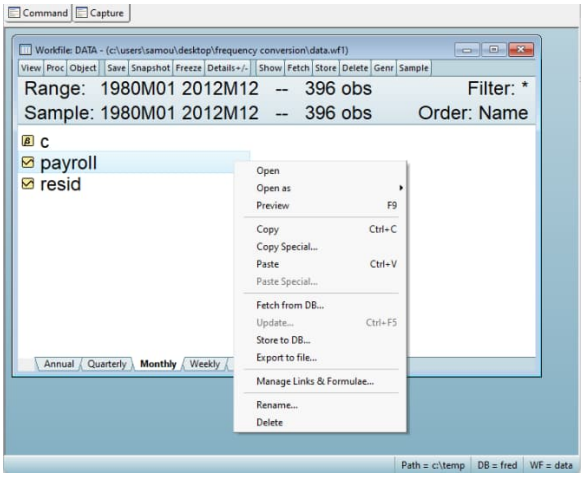
- It is just as easy in EViews to convert higher frequency data (e.g., monthly frequency) to lower frequency data (e.g., quarterly frequency).
- Suppose you have monthly *payroll* data and wish to convert to quarterly data.

Converting Monthly Frequency to Quarterly:

1. First, let's set the sample in the *Quarterly* page to include all observations. Click on the Quarterly page and type in the command window:

```
smpl @all
```

2. Now click on the *Monthly* page. Right-click on *payroll* series and select Copy.



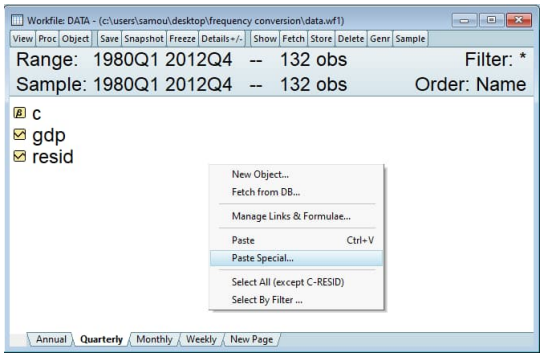
Frequency Conversion: High to Low

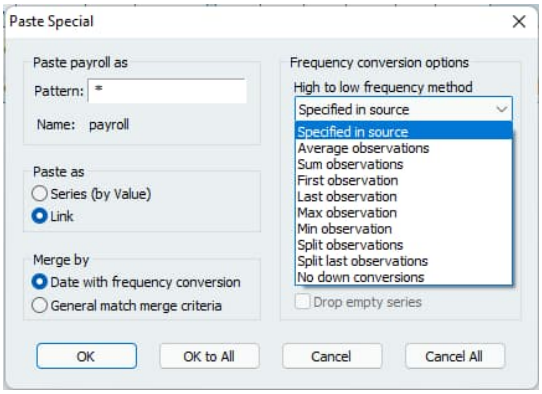
3. Click on *Quarterly* page (anywhere on the white workfile space).

4. Right-click and select Paste Special.

5. The Paste Special dialog box opens. On the left hand-side, select Link under the *Paste as* section.

6. Select Date with frequency conversion under *Merge by* section.

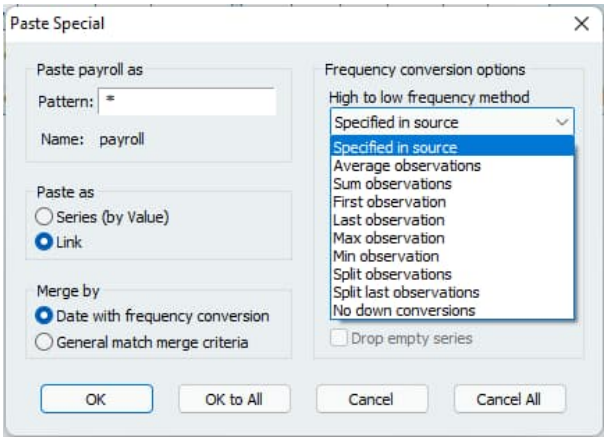




Method of Conversion: High to Low

7. Under "*Frequency Conversion Options"* select *High to Low.* There are several methods of conversion.

- ✓ ***Average Observations:*** Sets the low frequency data equal to the **average** of the observations in the corresponding high frequency data.
- ✓ ***Sum Observations:*** Sets the low frequency data equal to the **sum** of observations in the corresponding high frequency data.
- ✓ ***First Observation***: Sets the low frequency data equal to the **value of the first** observation in the corresponding high frequency data.
- ✓ ***Last Observation:*** Sets the low frequency data equal to the **value of the last observation** in the corresponding high frequency data.
- ✓ ***Max observation:*** Sets the low frequency data equal to the **maximum value** of the corresponding high frequency observations.



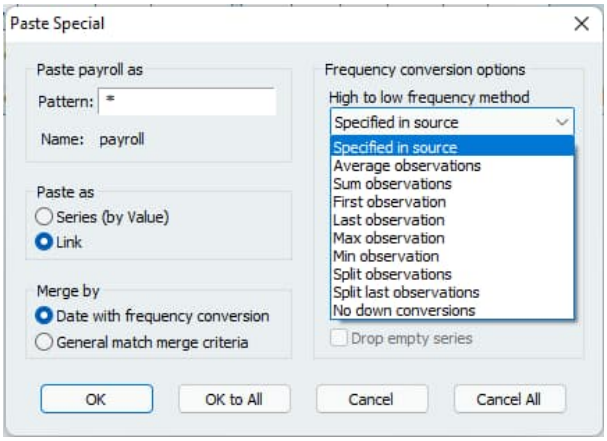
- ✓ ***Min Observation***: Sets the low frequency data equal to the **minimum value** of the corresponding high frequency observations.

Method of Conversion: High to Low

7. Under "*Frequency Conversion Options"* select *High to Low.* There are several methods of conversion.

- ✓ ***Split Observations***: *Creates m low frequency variables, where m is the number of high frequency observations in each low frequency period.* It sets the first low frequency variable equal to the first observation of the corresponding high frequency data; the second variable has the second observation, and so on for the mth high frequency value.

- ✓ ***Split Last Observation:*** *Creates m low frequency variables, where m is the number of high frequency observations in each low frequency period.* It sets the first low frequency variable equal to the ***last (***mth) observation of the corresponding high frequency data; the second variable has the (m-1)th observation, and so on for the first value.

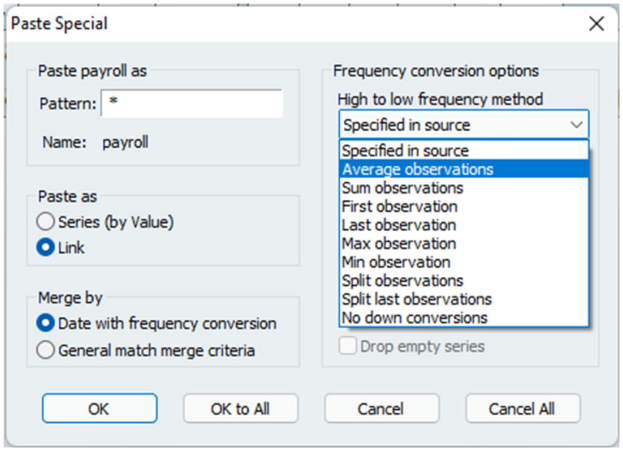


Frequency Conversion: High to Low Example 1: Average Observations

- *Average Observations:* Sets the low frequency data equal to the average of the observations in the corresponding high frequency data. Missing values are typically ignored.

High to Low Conversion: Example 1

1. Click on the *Monthly* page of the *Data.wf1* workfile.
2. Right-click on *payroll* series and select Copy.
3. Click on the *Quarterly* page, right-click and select Paste Special.
4. On the *Paste Special* dialog box, select Link under *Paste as* section.
5. Select Date with frequency conversion under *Merge by* section.
6. Select Average observations under "*High to low frequency method.*"
7. Click OK.

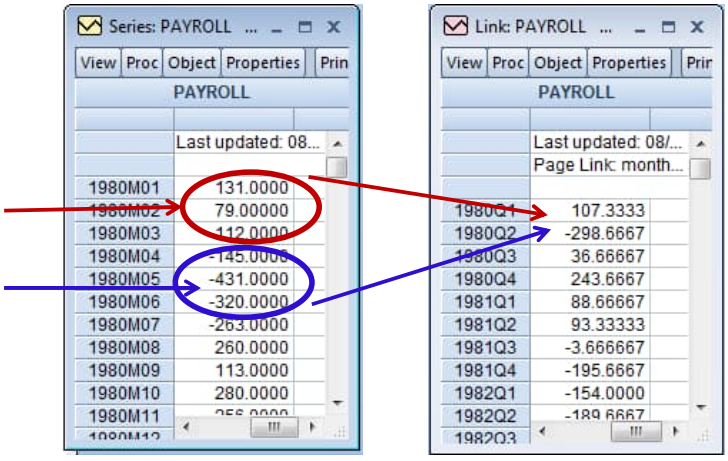


Frequency Conversion: High to Low Example 1: Average Observations (cont'd)

- Results for conversion by averaging are shown here.
- ✓ As you can see, this method fills in the observation of the low frequency data with the average of observations in the high page corresponding to that time period.
- ✓ For example, the average value for Jan 1980, Feb 1980 and March 1980 is 107.3333 and is placed in Q1 1980.

Average of three months = 107.33 (same as Q1 1980)

Average of three months = -298.66 (same as Q2 1980)

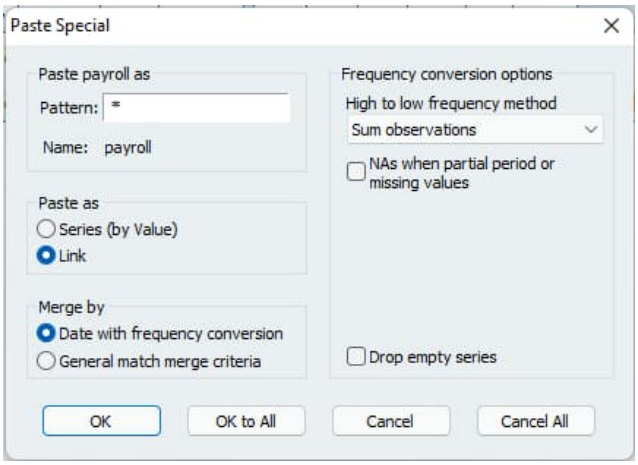


Frequency Conversion: High to Low Example 2: Sum Observations

- Sometimes it may make more sense for data to be summed up rather than averaged.
- If you have monthly employment data, for example, you might want to convert this to quarterly data by summing up net employment gains in each month of the quarter.

High to Low Conversion: Example 2

1. Click on the *Monthly* page of the *Data.wf1* workfile. Right-click on *payroll* series and select Copy.
2. Click on the *Quarterly* page, right-click and select Paste Special.
3. On the *Paste Special* dialog box, name this series *payroll1* (to not confuse it with the previous example).
4. Select Link under *Paste as* section.
5. Select Date with frequency conversion under *Merge by* section
6. Select Sum Observations under "*High to low frequency method.*"
7. Click OK.



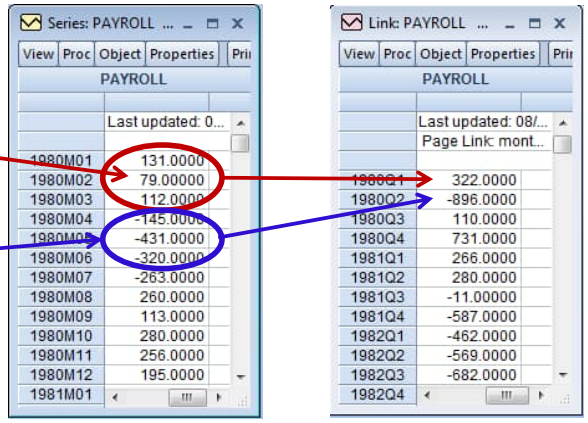
Missing values, NAs, are typically ignored.

Frequency Conversion: High to Low Example 2: Sum Observations (cont'd)

- Notice that Q1 1980 consists of the sum of observations for Jan, Feb and March of 1980.

Sum of three months = 322.00 (the same as Q1 1980)

Sum of three months = -896.00 (the same as Q2 1980)

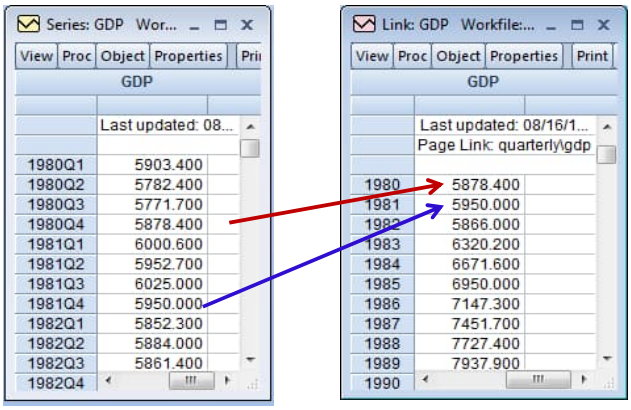


Frequency Conversion: High to Low Example 3: Last Observation

- In some cases, you might want to convert from a high to low frequency by preserving the last observation (or the first observation). This is a particularly useful feature for financial data.
- Suppose you want to convert quarterly *GDP* data into annual data by setting the value of the entire year equal to that of the last quarter:

High to Low Conversion: Example 3

1. Click on the *Quarterly* page of the *Data.wf1* workfile. Right-click on *GDP* series and select Copy.
2. Click on the *Annual* page, right-click and select Paste Special.
3. On the *Paste Special* dialog box, select Link under *Paste as* section.
4. Select Date with frequency conversion under *Merge by* section.
5. Select Last Observations under "*High to low frequency method.*"
6. Click OK.

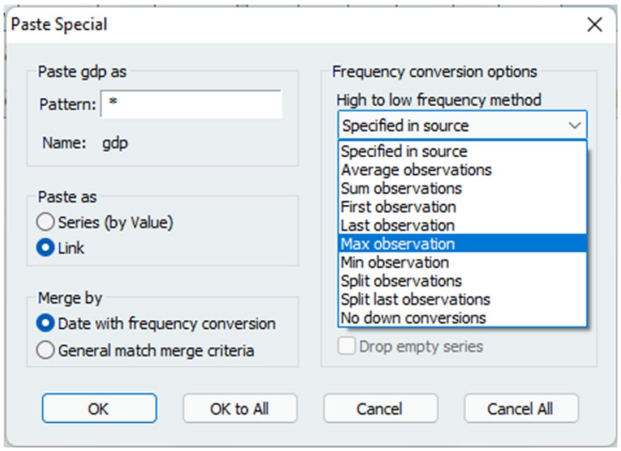


Frequency Conversion: High to Low Example 4: Max or Min Observation

- In some cases, you might want to convert high-to-low frequencies by selecting the maximum observation (or the minimum observation).
- Suppose you want to convert quarterly *GDP* data into annual data by setting the value of the entire year equal to the maximum (minimum) quarterly value during that year:

High to Low Conversion: Example 4

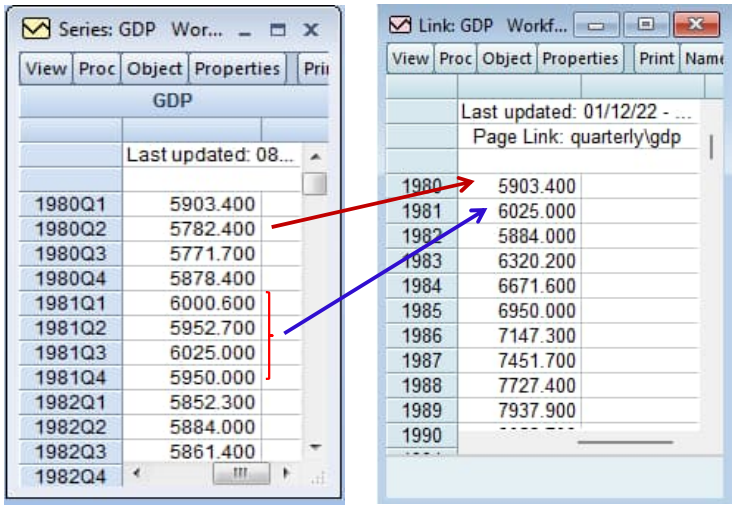
1. Click on the *Quarterly* page of the *Data.wf1* workfile. Right-click on *GDP* series and select Copy.
2. Click on the *Annual* page, right-click and select Paste Special.
3. On the *Paste Special* dialog box, select Link under *Paste as* section.
4. Select Date with frequency conversion under *Merge by* section.
5. Select Max (or Min) Observation under "*High to low frequency method.*"
6. Click OK.



Frequency Conversion: High to Low Example 4: Max or Min Observation

Maximum value of GDP during 1980 was 5903.4 (i.e., the 1980Q1 value). The 1980 value for GDP would be 5771.7 if the minimum value was used to convert GDP to the annual frequency.

Maximum value of GDP during 1981 was 6025.0 (i.e., the 1981Q3 value). The 1981 value for GDP would be 5950.0 if the minimum value was used to convert GDP to the annual frequency.



Frequency Conversion: Split Observations

We can also create lower frequency series from higher frequency series by 'splitting' the values in the higher frequency series into distinct lower frequency series. This is called 'split' or 'skip' sampling, and especially useful for nowcasting regressions involving variables with mixed frequencies.

Skip sampling involves (a) selecting a starting observation in the higher frequency series and (b) selecting every m^{th} observation in the higher frequency series to build the lower frequency series.

For example, for the monthly payroll data, one can derive a quarterly payroll series by selecting, as its initial value, the 1980M1 value (= 131.0), and then the 4th value (-148), followed by the 7th value (-263.0) and so on till the last available value of the high frequency series is reached.

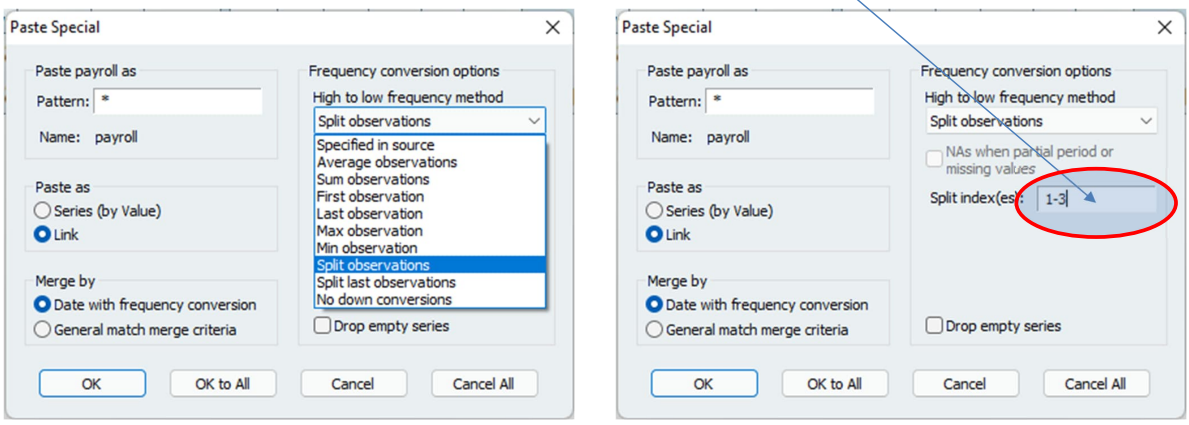
Another quarterly series for payroll could be derived by starting with the 2nd value of payroll (1980M2 = 79.0) and repeating the above procedure.

Lastly, we could start with the 3rd value, 1980M3, to obtain a third (lower frequency) series.

Frequency Conversion: Split Sampling

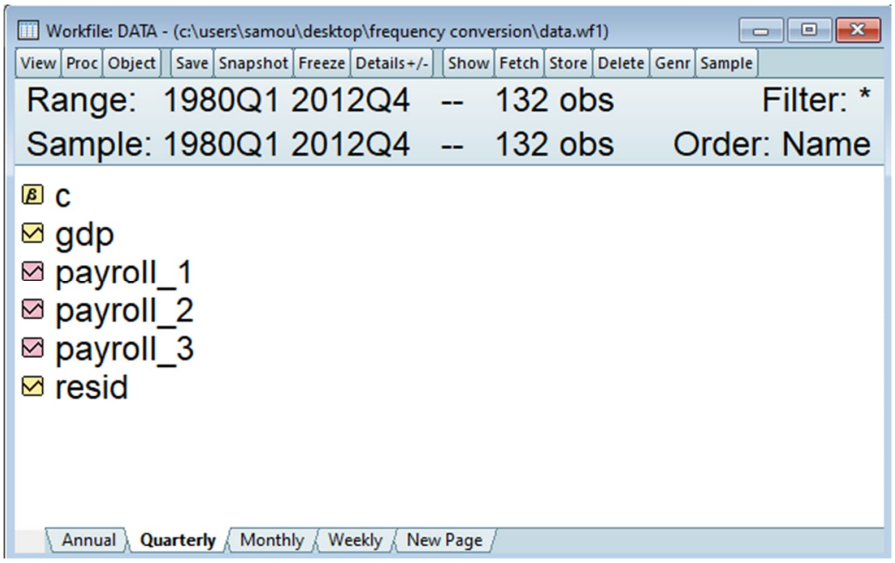
Split indexes, or series name tags, need to be assigned to differentiate the low frequency series derived from the high frequency series.

In this case, split indexes of “1-3” have been assigned.



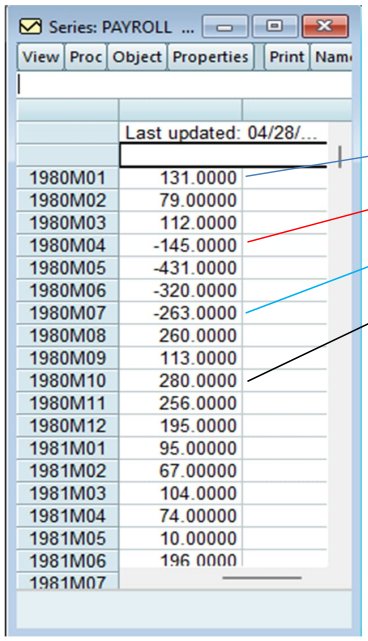
Frequency Conversion: Split Observations

Notice that three quarterly series have been created from the monthly payroll data. The user assigned indexes are used to name the series.

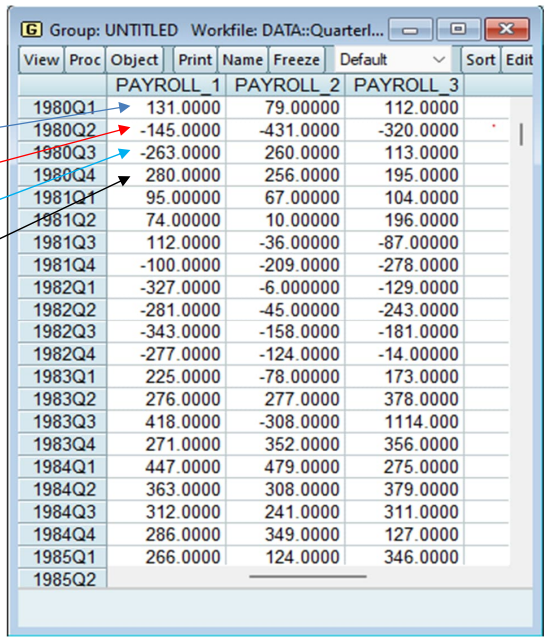


Frequency Conversion: Split Observations

Monthly Payroll

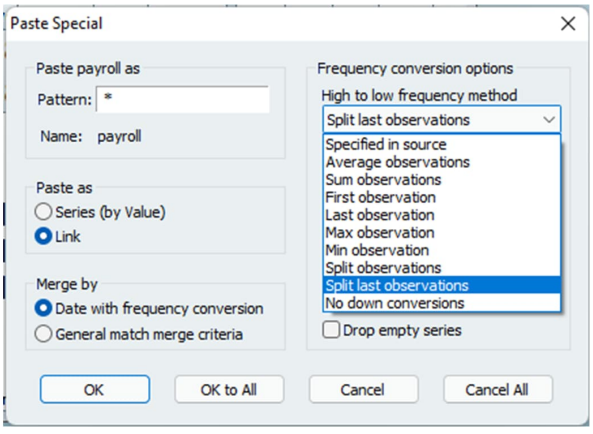


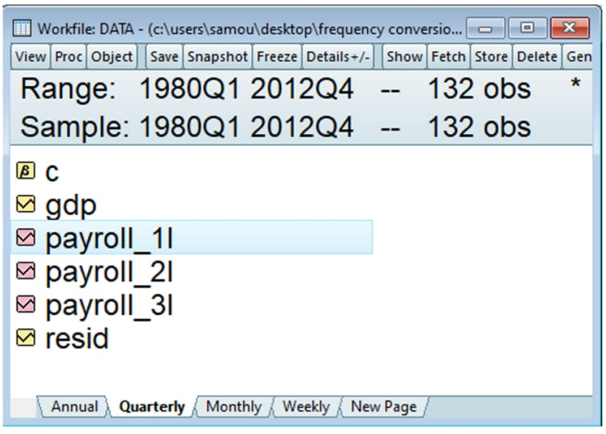
3 Quarterly Series: Split Sampled



Frequency Conversion: Split Last Observation

For split sampling using the last observation, the same procedure is used except that the first series starts with the last (in this case) monthly observation of the first quarter.

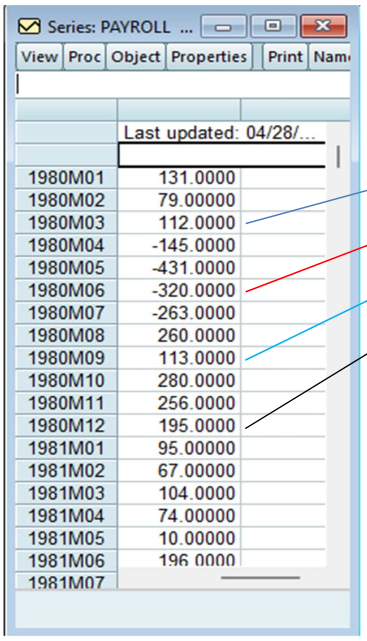




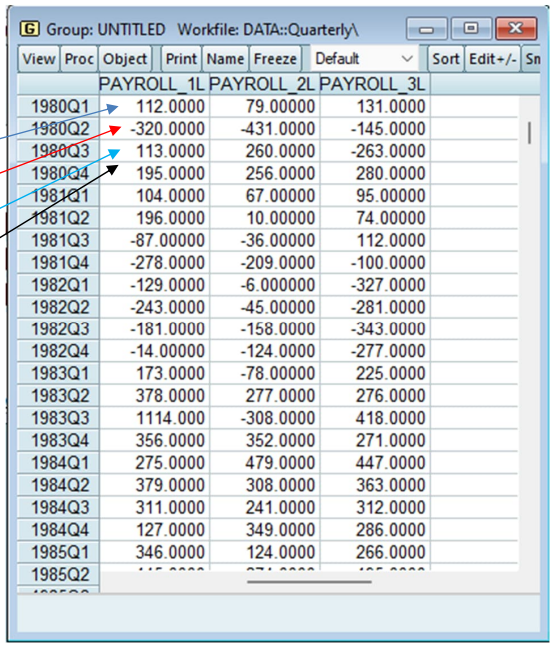
Notice that the resulting series are tagged with an 'l' at the end of their name to indicate that the split sampling started with the last monthly observation rather than the first.

Frequency Conversion: Split Last Observation

Monthly Payroll



3 Quarterly Series: Split Last Sampled

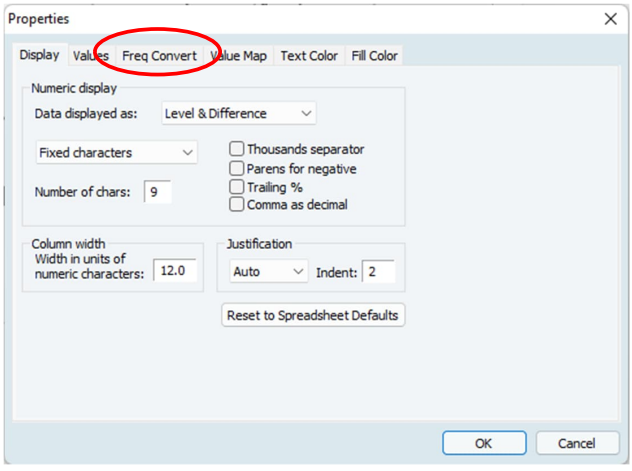


Specifying Default Conversion Methods

It would be tedious to carry out copy/paste operations and have specify the various options each time a frequency conversion is required.

When performing frequency conversion of one or more series, EViews can refer to the attributes of the series to perform the conversion. These settings may be specified using the Freq Convert tab of the Properties dialog of the series (accessible from Spreadsheet View).

Click on the Properties button on the series toolbar and select the Freq Convert tab to access the dialog box.



Specifying Default Conversion Methods

The series properties dialog box allow you specify the default conversion methods to be used when moving from high to low (e.g., Average) frequency or from low to high (e.g., Linear)

This enables you to do frequency conversion by copying and pasting series from one pagefile to another with a different frequency.

